

Westley Kleinman

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EDUCATION

Duke University

Bachelor of Science, Mechanical Engineering

May 2028 (Expected)

GPA: 3.88

EXPERIENCE

Re:3d

Research and Development Engineering Intern

May 2026 – August 2026

Austin, Texas

- Developed an end-to-end material reclamation process converting M61 gas filter canisters (military base waste) into 3D-printable rPP feedstock — removing adhesives/contaminants, then granulating for FGF pellet printing — and used it to print drone frames, completing initial proof-of-concept testing for real-world military applications using FGF style printing.
- Observed rPP printing behavior on the reclaimed material and iterated the drone frame's design for manufacturability (DFM), validating structural performance via SolidWorks FEA (FOS ~3.4x at max throttle) while minimizing weight (90 grams).
- Diagnosed and repaired hardware faults (Klipper MCU errors, stepper driver failures) to restore broken and unused printers to production with upgraded components, designing test parts in SolidWorks, including adding two pieces to production.
- Developed FGF print profiles for new materials, including a LFAM producing military pallet replacements (SBIR Grant), and completed contract print jobs for customers.

Time Bicycles

Process Optimization and Test Engineering Intern

May 2025 – August 2025

Greenville, SC

- Overhauled RTM mold geometry and machine calibration for carbon-rim fabrication, eliminating dry zones and standardizing resin distribution while maintaining competitive weight.
- Conceived and built a custom wheel-impact rig (winch + electromagnetic stopper), validating rim durability to 40 J UCI standard with capability up to 200 J.

PROJECTS

Monte

Founder

2021 - Present

[Monte3D.com](https://monte3d.com)

- Engineered a software pipeline (Python, GDAL/geopandas, Blender) converting a mountain name into ready-to-print STL files, automating GIS extraction and mesh generation, then calibrated multi-material print profiles (flow ratio, pressure advance, infill) for dimensional accuracy across elevation bands.
- Founded and built the company from concept to production, developing the full-stack DTC platform (Next.js, TypeScript, Three.js) and securing a provisional patent and distribution partnerships that drove four-figure revenue.

Automatic Bucket Washing System

Lead Test Engineer

Fall 2024

EGR101 Freshman Design

- Collaborated on a 4-person team to design and fabricate a two-stage cleaning system for CompostNow (client), combining a spray-nozzle rinsing device with a gear-driven brushing device (24V motor, 3D-printed PLA helical gears), reducing cleaning time by 43% and reduced users from 4 to 1.
- Modeled components in SolidWorks and fabricated via laser cutting and 3D printing; validated performance through structured testing (cleanliness, odor) against defined design criteria across a 20-bucket sample.

LEADERSHIP/ACHIEVEMENTS

Duke Club Cycling

President/Lead Safety Officer

August 2025 - August 2026

- Lead executive planning and safety operations for the club, prioritizing development pathways for entry-level riders; built DukeCycling.dev, a full-stack site for ride scheduling, team history, and ride proposals

MaccabiUSA Pan American Athlete

Road Cycling

December 2024 - January 2025

- Selected as 1 of 3 athletes to represent MaccabiUSA in cycling at the Pan-American Games in Buenos Aires, Argentina

Cyberpatriot

Team Founder, 3x State Champion

August 2021 - May 2024

- Founded and led a high school cybersecurity competition program, building a training curriculum for 15+ members across 3 teams and winning state championships in 2021, 2023, and 2024